
DATA STRUCTURES AND ALGORITHMS

An Assignment Submission

1. Using a programming language of your choice, in groups, write code to represent each of the data structure classification and types.

- Data structure can be defined as the group of data elements which provides an efficient way of storing and organizing data in a computer so that it can be used efficiently.

2. Research where the data structures types are applied and give reasons

Arrays:

Application:

- ✓ Image processing (e.g storing pixel values)-enabling efficient manipulation and filtering
- ✓ Spreadsheet data storage
- ✓ Storing marks of students
- ✓ Storing list of phone contacts

Reason:

-This is because arrays offer fast random access, making them ideal for situations where data is accessed by index.

Linked list:

Application:

- ✓ Music playlist
- ✓ Dynamic memory allocation
- ✓ Database query results

Reason:

- This is because linked list allow efficient insertion/deletion of elements without shifting entire data

Stack:

Application;

- ✓ Function calls
- ✓ Undo/redo functionality in text editors
- ✓ Browser history

Reason;

- Stacks follows Last In, First Out (LIFO) execution making it perfect for tracking sequential actions and keep track of most recent action.

Queue:

Application;

- ✓ CPU scheduling
- ✓ Printer job scheduling
- ✓ Network buffers

Reason;

- Queue follows first in first out principal ensuring for orderly processing and fairness

Tree:

Application;

- ✓ File systems
- ✓ Database indexing
- ✓ Organization charts

Reasons;

- Shows hierarchical relationship enabling fast search, insert and delete operations, especially when balanced.

Graphs:

Application;

- ✓ Social networks
- ✓ Google maps in routes
- ✓ Network connection

Reason;

- It shows relationship between objects enabling efficient traversal and analysis.

3. Give examples of applications that are using the data structure type and algorithm. Give reasons why

1. Google search engine

- It applies the trees, graphs and hash tables. This is because the hash tables allow very first lookup of web pages, the trees get to organize billions of pages for quick searching and the graphs represents links between website to rank importance

2. WhatsApp

- It applies queue and linked list as data structures and the message scheduling and encryption algorithms as algorithms. This is because queue gets to ensure

messages are delivered in order (FIFO), linked list handles dynamic chat history

3. Online shopping (Jumia, amazon)

- The online shopping applies graphs and stacks data structures ; recommendation algorithms and sorting and searching algorithms. The graphs link users to the product they like. The stacks manage back navigation and search history.

4. Operating systems

- The operating system applies the queue, stack and linked list data structures together with the CPU scheduling and memory management algorithms. The queue gets to schedule the processes fairly, the stack manages the function calls while the linked list manages free memory blocks.

5. Web browser

- It applies the stack data structure which is the backtracking algorithm(FIFO). The stack manages browser history, allowing user to navigate back and forth.

6. Youtube

- It applies the queue and the graph where the graphs represent video relationship and collaborative filtering recommends videos based on users. The queue manages video buffering ensuring smooth playbacks.

7. Google maps and gps

- It applies the graph data structure since roads are naturally represented by graphs hence determines the shortest and most efficient routes

8. Facebook

- It applies the graphs which represent the friendship, the hash tables which store the user profiles for quick access . the friendship suggestion and news feed ranking are the algorithms which help analyze relationships.

4. Research how data structures and algorithms work within systems

1. Operating system

- In the operating system the stack manages function execution, the queue manages processes or the CPU scheduling, the tree represents file directories, the linked list the memory allocation and the algorithms processes scheduling.

2. Storage and organization

- System handles large number of information such as files. The arrays get to store fixed records like students marks, the linked list allow dynamic data such as playlist or contact list, trees organize hierarchical data like the file folders and the graphs represent network such as the roads or socials.

3. Memory management

- Operating systems rely on data structures to manage RAM. The free memory blocks are stored using linked lists, the function calls are controlled using a

- stacks and the processes waiting for CPU uses a queue. This ensures fair and efficient use of hardware resources.
4. Banking system
 - In banking systems the trees account the indexing the hash tables enables fast login into the bank system and the algorithms helps in transaction validation.
 5. Decision making systems
 - Decision making systems rely on algorithms for intelligent decisions. The GPS uses the shortest path algorithms, online stores uses recommendation algorithms and the operating system use scheduling algorithms.
 6. Data communication in networks
 - Data structures control movement of data. The queue holds packets in routers, the stacks manage browsers navigation and the hash tables locate user sessions quickly.

Reference; Mount Kenya university. (2024). Data structures and algorithms course module.

Project github link;

<https://github.com/davine200/data-structures-and-algorithms.git>